# Audio

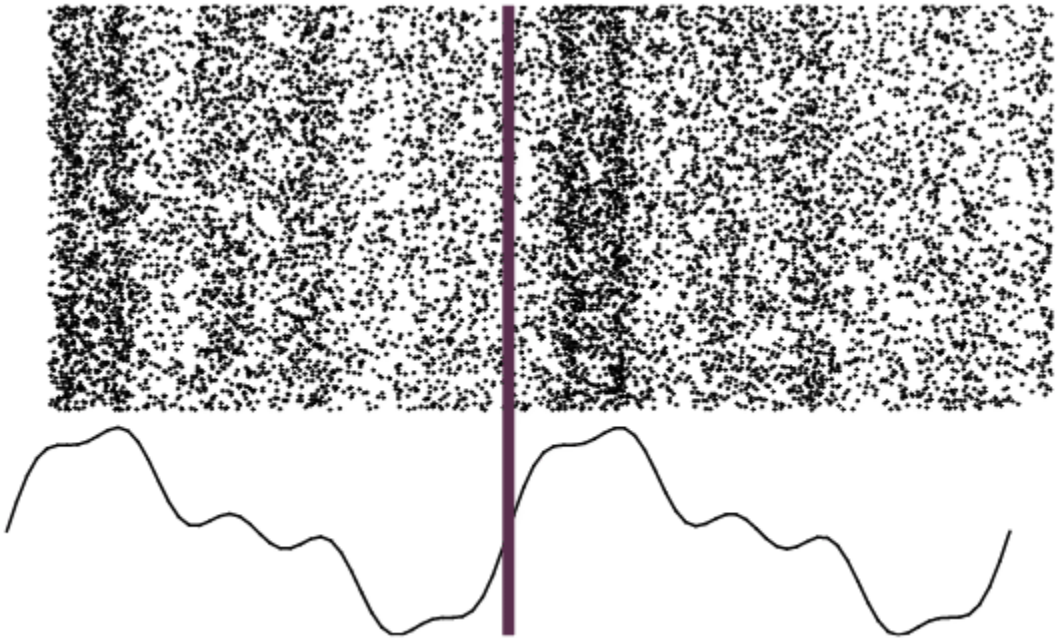
Dr. Josef Fruehwald

## The Upshot

Recording sound and doing acoustic analysis is all about managing trade offs.

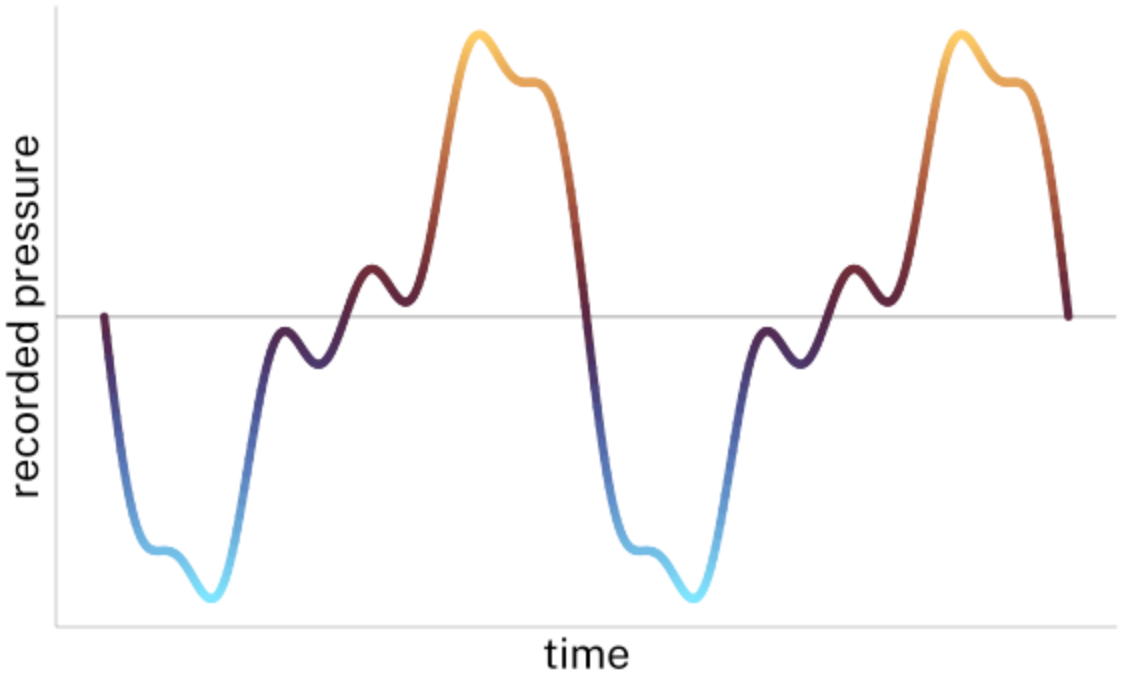
## Sound

From space:

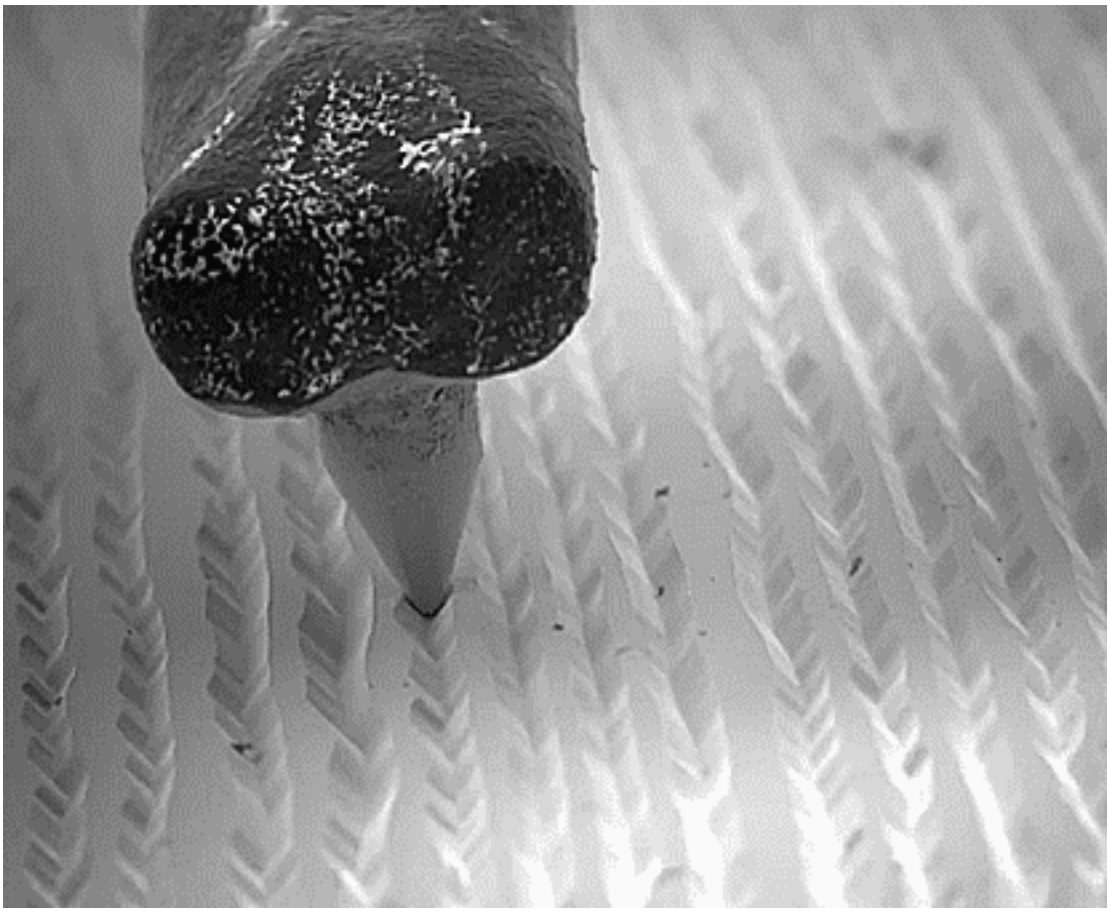


## Sound

To time:


recorded pressure
time

## Analog Audio



## Analog Audio








## Analog Audio

YouTube

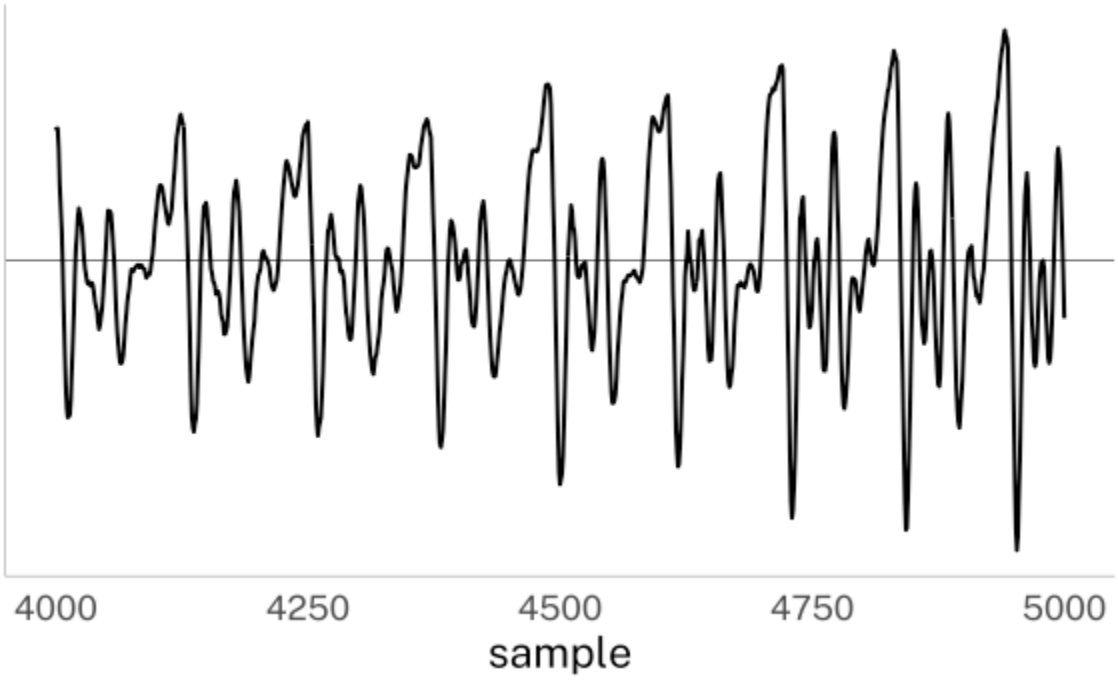


## Digital Audio

Instead of etching continuous grooves, we can note down a sequence of high and low pressure values.

```
library(tuneR)
wav <- readWave(here::here("data/audio/harmonics.wav"))
wav@left[4000:4010]
```

```
 [1]  4640  4567  4592  3869  2631  1848  1090  -239 -1731 -3085 -4311
```




## Digital Audio

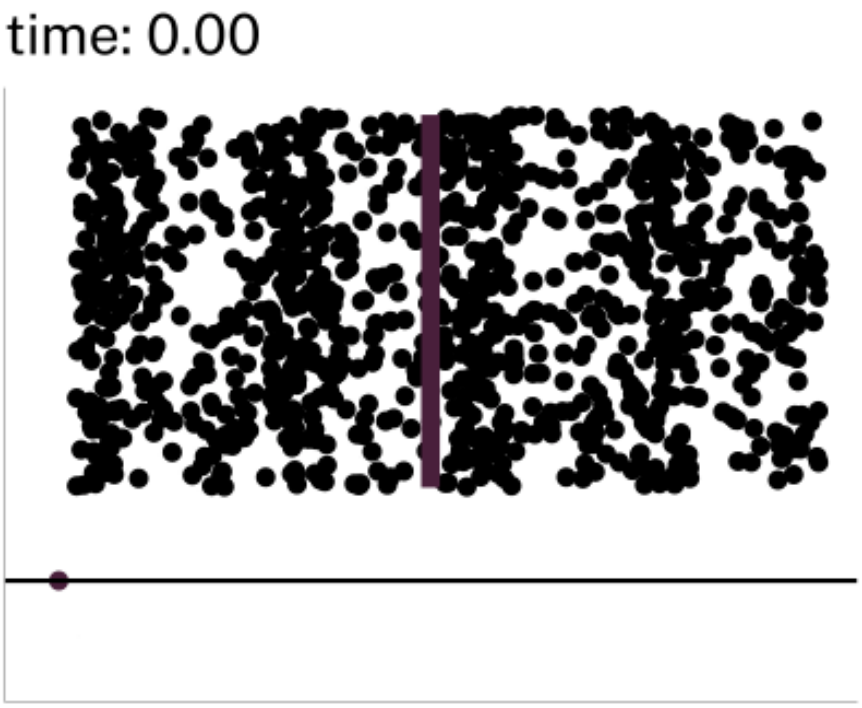
Let's say we decide to note down that the air pressure at the first moment in the recording was −194.

*When* do we decide that the *second* moment is, to write down the *second* air pressure?

*Time* doesn't come in discrete moments.

## Digital Audio

Sampling rate




## Sampling Rate

- Described in Hz.
- 16,000 Hz sampling rate = 16,000 samples per second

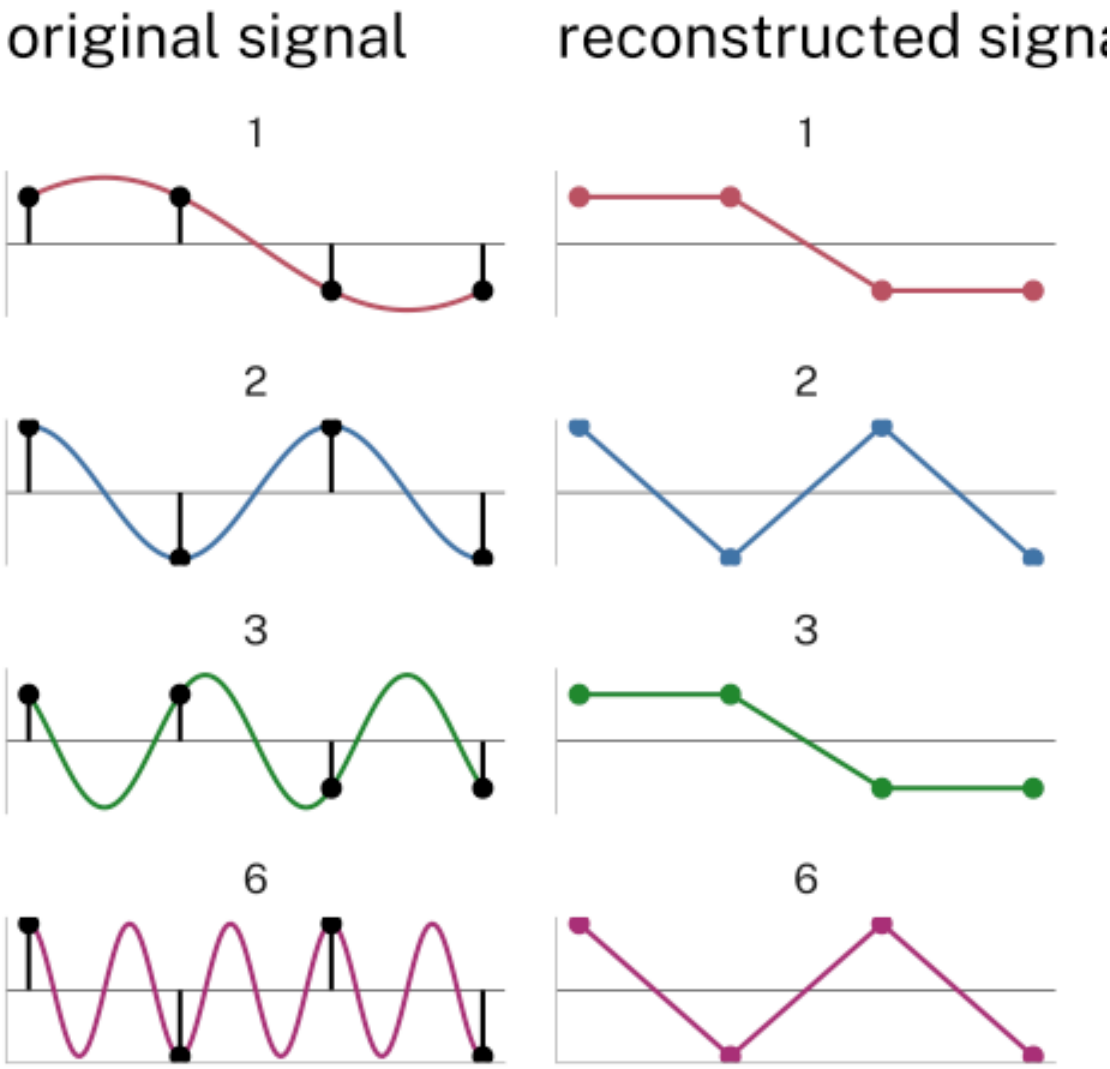
## Sampling Rate

Common Sampling rates

| Rate | Context |
|---|---|
| 8,000 Hz | The sampling rate on telephones. |
| 16,000 Hz | Commonly used in speech processing. |
| 44,100 Hz | Standard sampling rate for MP3s |
| 48,000 Hz | Standard sampling rate for professional digital recording |

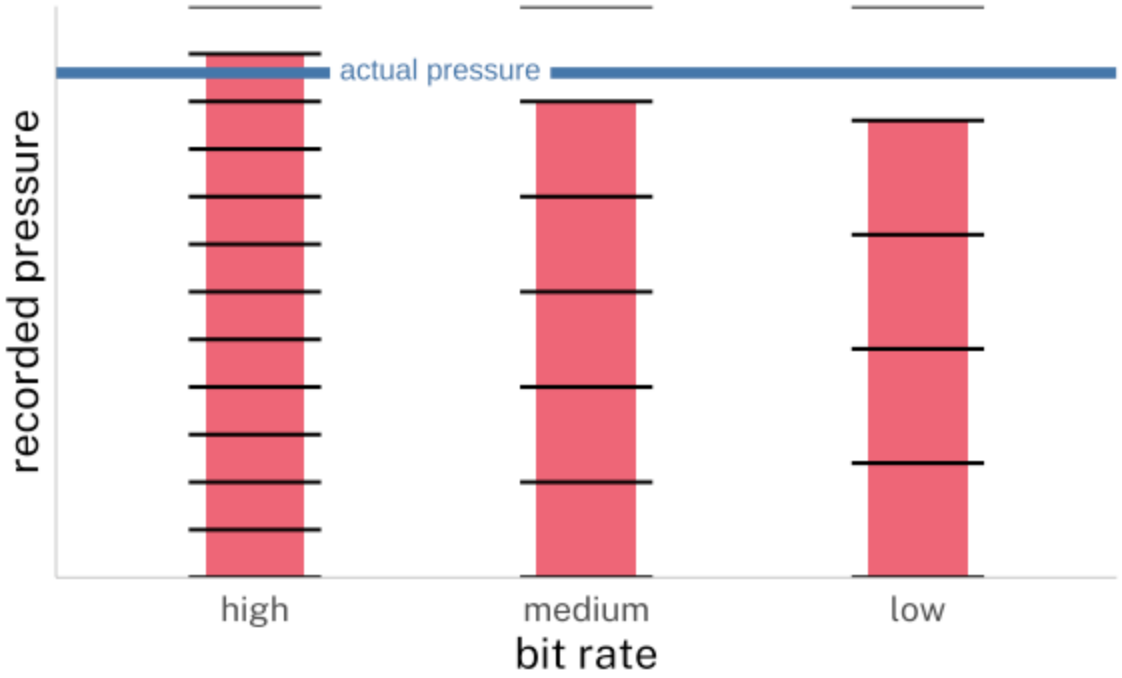
## Sampling Rate

You can record sound frequencies up to $\frac{1}{2}$ the sampling rate (the Nyquist Frequency).




## Bit Rate

Computers can only record numbers with finite precision.




## Sample and Bit Rate

Why not always max out sampling rate and bit rate?

## Precision & File Size

**Duration**
0.89s

**Sampling Rate**
44100 Hz

**File Size**
77.1K

**If it was 1 hour**
305M

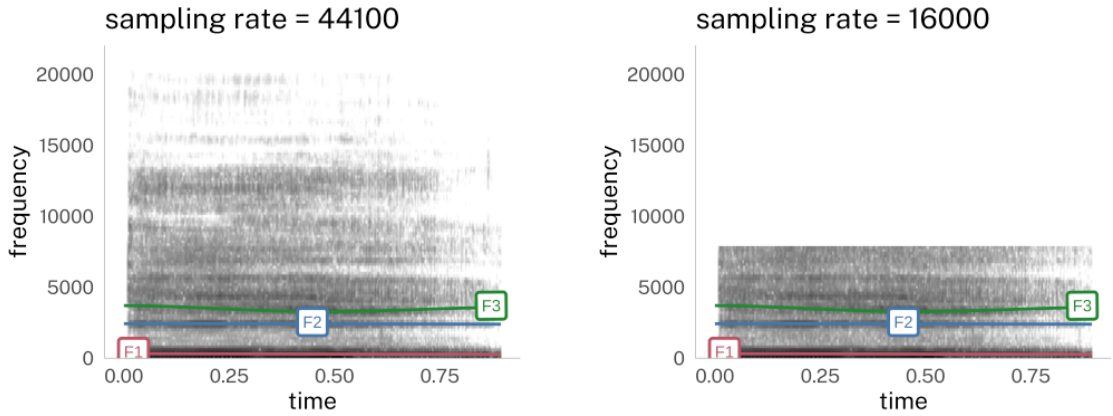
**Duration**
0.89s

**Sampling Rate**
16000 Hz

**File Size**
28K

**If it was 1 hour**
111M

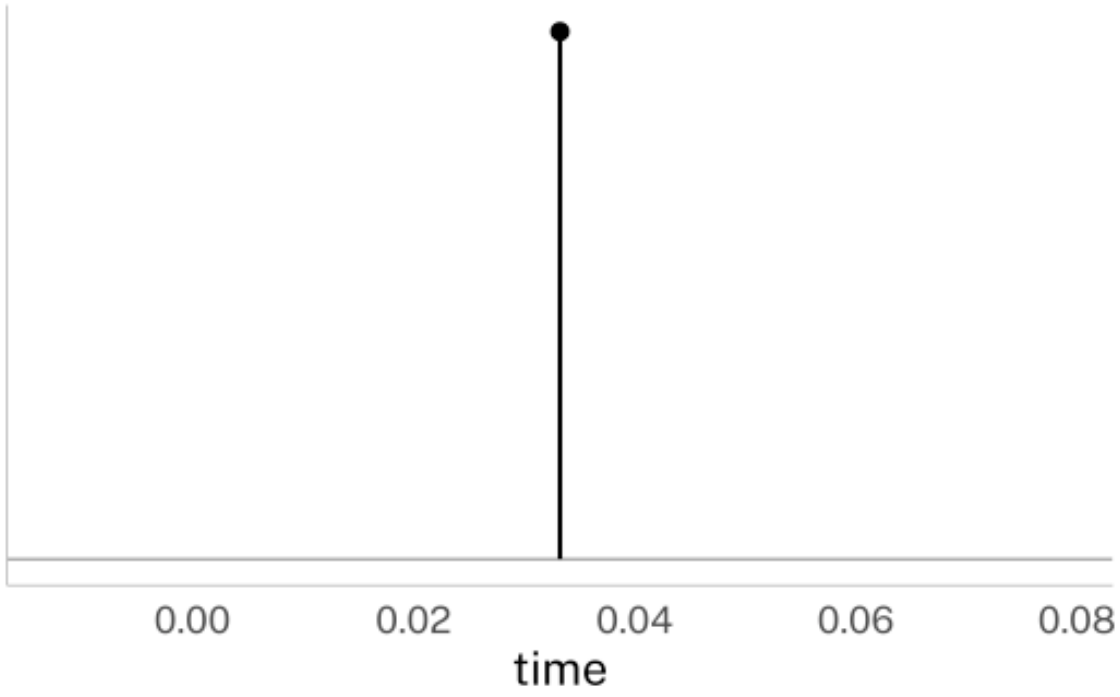
## What's really necessary?




## Acoustic analysis

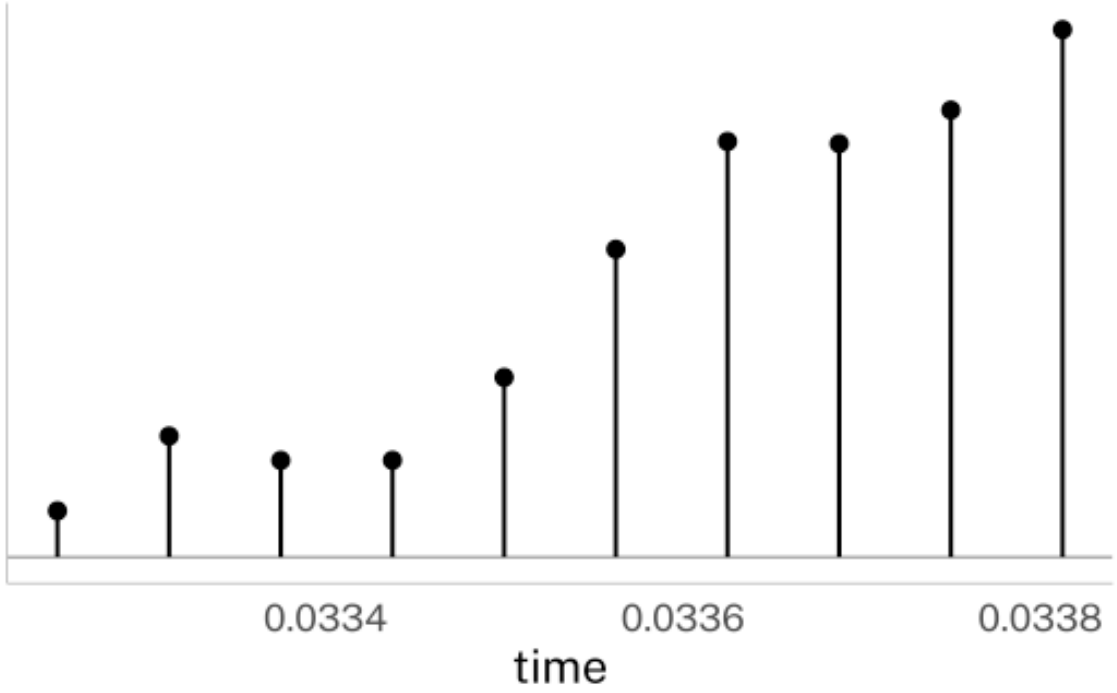
Let's say you've decided on a sampling rate, and now you want to estimate the fundamental frequency.

At a single point in time, we have a single sample. At what frequency is the signal alternating?



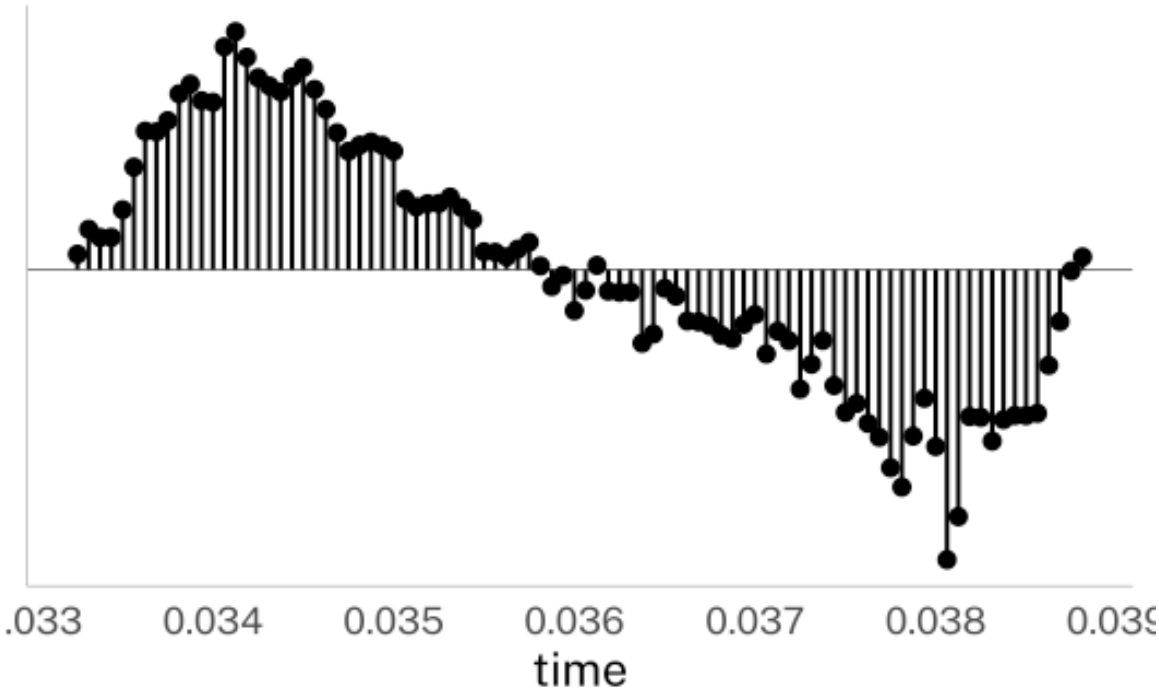
## Acoustic Analysis

Ok, here's 10 points.




## Acoustic Analysis

It looks like we need 90 samples or $\frac{90}{16000}$ = 0.005625 seconds to get a single period.

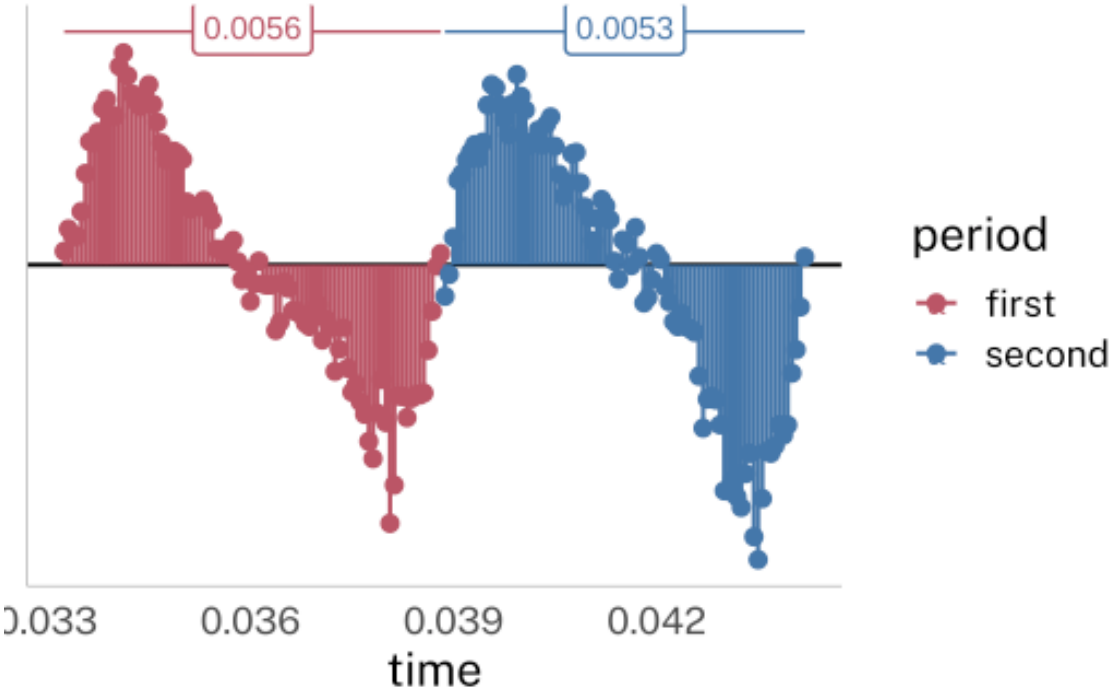



## Acoustic Analysis

Analyses of frequency can only be done over spans of time. We always need to choose an analysis window size.

## Acoustic Analysis

The second period isn't exactly the same.




## Acoustic Analysis

| period | samples | duration (s) | frequency (Hz) |
|---|---|---|---|
| first | 90 | 0.00562 | 178 |
| second | 86 | 0.00537 | 186 |

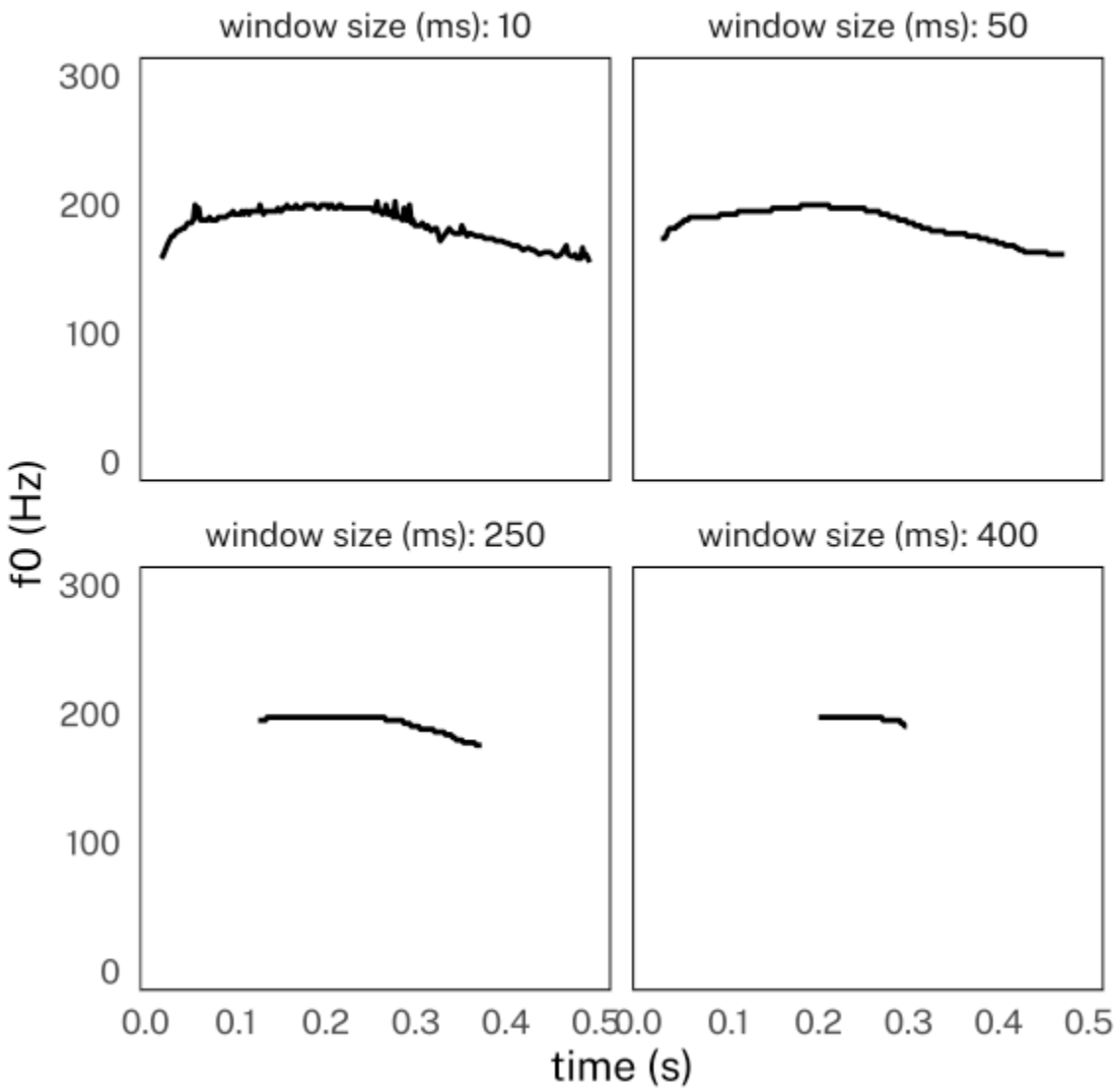
## Acoustic Analysis

What if we expanded the analysis window to the entire vowel for greater accuracy?

| time (s) | frequency (Hz) |
|---|---|
| 0.447 | 176 |

But... the way F0 changed over the course of the vowel has been lost.

## Acoustic Analysis

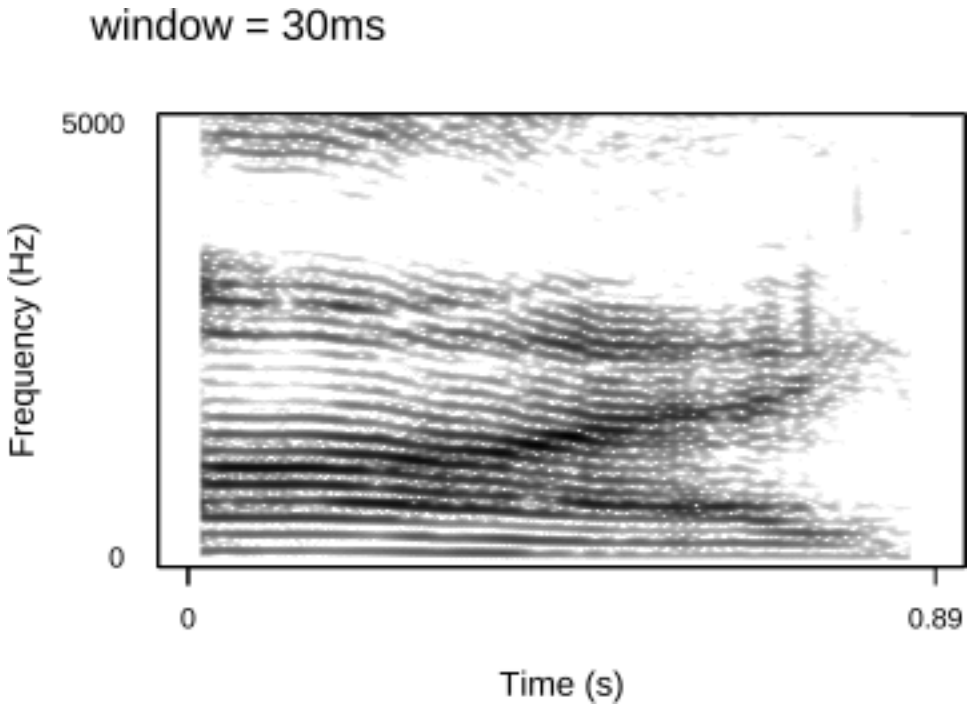



## Acoustic Analysis

These same trade off between time and frequency needs to be made with spectrograms as well.

## Acoustic Analysis

Narrow band spectrogram = wide analysis window




## Acoustic Analysis

Wide band spectrogram = narrow window

